Jeffery Ye

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Software and AI engineer with experience designing, developing, and deploying full-stack systems (React, Django, SQL/NoSQL) and agentic AI solutions (LangChain, PydanticAI).

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, R
AI/ML: Agentic AI, LangGraph, LangChain, PydanticAI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LLM Evaluation, Natural Language Processing (NLP), PyTorch
Full-Stack: ReactJS, NodeJS, SQL (Postgres, MySQL), NoSQL (Neo4j, pgvector), Flask, FastAPI, Django
Platforms & DevOps: AWS, Spark, Databricks, GCP, Docker, Azure, Git, CI/CD, Linux/Unix, Hadoop

EDUCATION

BS Informatics (Focus: Software & AI) | University of Washington | Expected June 2026

- **Relevant Courses:** Data Structures/Algorithms, Adv. Databases, Web Dev., Adv. Data Science
- **GPA**: 3.55

PROFESSIONAL EXPERIENCE

Software/AI Research Assistant | Seattle Children's Hospital | April 2025 - Present

- Designed, developed, and deployed two multi-agent AI systems to automate complex research workflows, delivering tools that **reduced experimental design time by 95%**
- Took the initiative by **leading** brainstorming/design meetings, **interviewing** users, and shadowing stakeholders to iteratively improve upon the applications I developed.
- Engineered a scalable data pipeline integrating SQL, graph (Neo4j), and vector (pgvector) databases to synthesize insights from unstructured literature, genomic, and proteomic data.
- Developed secure full-stack applications with Django, Flask, and FastAPI for researchers.

Teaching Assistant (Informatics) | University of Washington | March 2025 - Present

- Delivered weekly hour-long lectures on Informatics concepts (e.g., AI/ML, data science) to 50+ students, achieving a **4.9/5.0 rating** (Top 20% across all University of Washington instructors).
- Designed, oversaw, and administered LLM and ML based activities to a class of 200 students

PROJECTS AND PUBLICATIONS

Automated extraction and optimization of protein purification protocols using multi-agent large language models by Jeffery Ye et al. | bioRxiv (Preprint) | 2026

- Led the design, development, and evaluation of the full-stack application, coordinating with domain experts to iterate on designs and validate outputs.

AI Expense Categorizer | 2025

- Ideated and developed a fintech AI agent with PydanticAI to automate the secure classification of unstructured financial data, completing **~2 hours of manual work in 30 seconds**.

Musicalendar | 2025

- Developed a full-stack web application using ReactJS and a Flask back-end, deployed on AWS.
- Engineered a max-flow scheduling algorithm to optimize high-volume, 1-on-1 meeting bookings, replacing inefficient manual coordination via email chains and spreadsheets.

Educational Research STM | 2024

- Applied structural topic modeling to analyze **800+ educational policy research papers**, developing expertise in NLP pipelines, Spark, and statistical validation of topic model outputs.

Optum Claims Analysis | 2023

- Engineered SQL queries with JDBC to analyze 50k+ urinary cancer records in the Optum dataset.
- Synthesized findings into reports on treatment costs, utilizing Python (Pandas) for validation.